



1. รายละเอียดของโจทย์

1.1 ชื่อโจทย์ (ต้องเป็นภาษาอังกฤษ)

Pieces of the Past

1.2 หมวดของโจทย์ (Red Team, Blue Team, Miscellaneous)

Miscellaneous

1.3 ชื่อทีม

0xBl4ck-c4t-t

1.4 ระดับการแข่งขัน (มัธยมศึกษา, อุดมศึกษา)

อุดมศึกษา

1.5 รายชื่อสมาชิกในทีม

หัวหน้าทีม: -

สมาชิกคนที่ 1: -

สมาชิกคนที่ 2: -

2. รายละเอียด Write-up และการแสดงขั้นตอนการแก้ปัญหาจากโจทย์

RECOVERED ARTIFACT #03

16 1 19 20 5

DATA:

3633323328352f276906090316060d15131c1711163a3e65363
31e2b64370f4b5935372428353526247c3d0310650d3464175a
4b3d313d247b732424233536362c3e

A begins with 1.

Z ends with 26.

What you find may be the key.

Solution:

So It said that A begins with 1 and Z ends with 26, I thought of A1Z26 and put it through the decoder and we got PASTE.

A1Z26 substitution

The screenshot shows a web interface for a 'NUMBER TO LETTER A1Z26 CONVERTER'. On the left, under 'Showing most likely results', there are two options: '[A1Z26] PASTE' and '[A0B1Z25] QBTUF'. The main area has a title 'NUMBER TO LETTER A1Z26 CONVERTER' with a refresh icon. Below it is a section '★ LETTER TO ALPHABET NUMBER A1Z26 CIPHERTEXT (NUMBERS)' containing a text input field with the value '16 1 19 20 5'. Below the input field is a checkbox labeled '★ TRY ALTERNATIVE ALPHABETS (SHIFTED, REVERSED)' which is currently unchecked. At the bottom right is a green button labeled '▶ DECRYPT AUTOMATICALLY'.

Now we will do the XOR Decoding.

The key is PASTE.

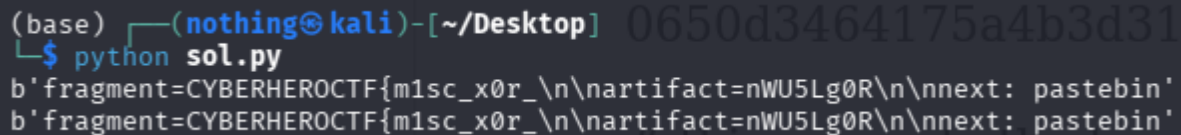
```
import binascii
```

```
hex_data =  
"3633323328352f276906090316060d15131c1711163a3e6536  
331e2b64370f4b5935372428353526247c3d0310650d3464175  
a4b3d313d247b732424233536362c3e"
```

```
# this converts the hex to byte  
data = bytes.fromhex(hex_data)  
key = b"PASTE"
```

```
# this xor the the data that we get to the get first part of the flag  
decoded = bytes(  
    b ^ key[i % len(key)]  
    for i, b in enumerate(data)  
)
```

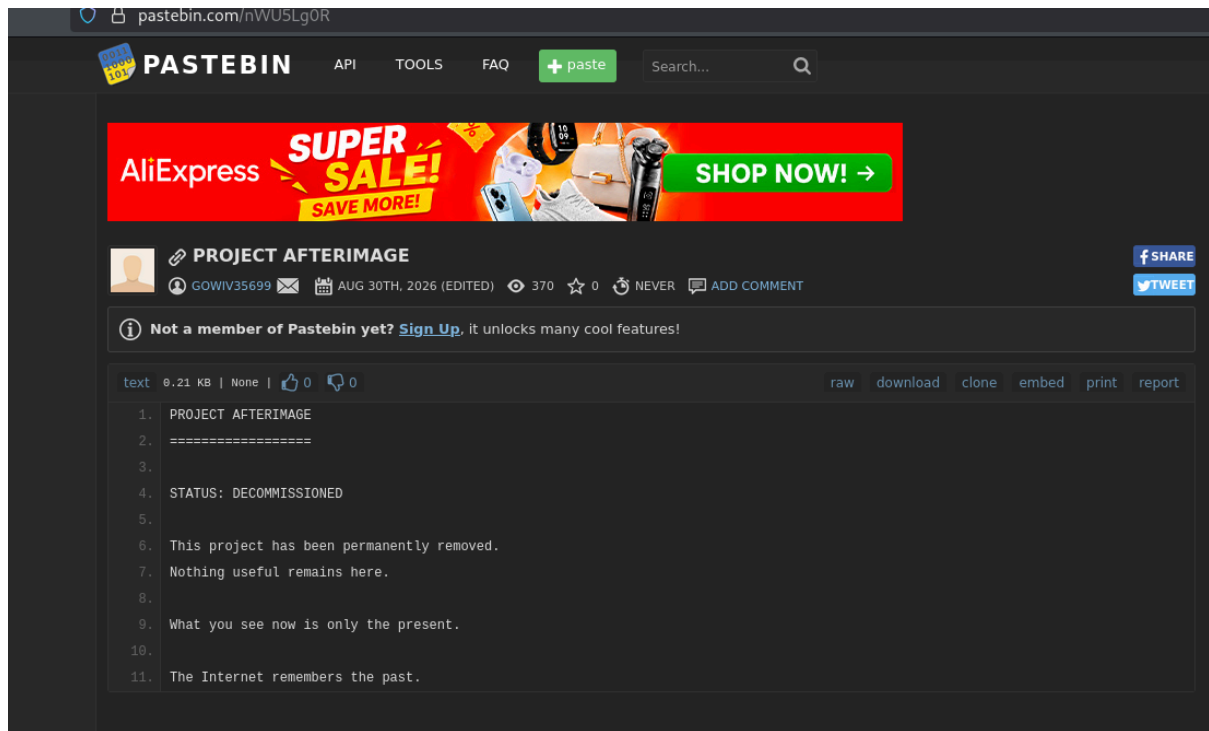
```
print(repr(decoded))  
print(decoded)
```



```
(base) [nothing@kali] - [~/Desktop] 0650d3464175a4b3d313  
$ python sol.py  
b'fragment=CYBERHEROCTF{m1sc_x0r_\n\nartifact=nWU5Lg0R\n\nnext: pastebin'  
b'fragment=CYBERHEROCTF{m1sc_x0r_\n\nartifact=nWU5Lg0R\n\nnext: pastebin'
```

First flag part = CYBERHEROCTF{m1sc_x0r_
Artifact = nWU5Lg0R
Next = pastebin

So, next I went to the pastebin website to see what's on that



PROJECT AFTERIMAGE

=====

STATUS: DECOMMISSIONED

This project has been permanently removed.
Nothing useful remains here.

What you see now is only the present.

The Internet remembers the past.

This show the current edited but we have a way to check the
past edit by using **wayback machine**

PROJECT AFTERIMAGE

Build: 0.9.3

Status: Migration Pending

Recovery Fragment:

p4st3_n3v3r_

ARCHIVE DATA:

27762622613d3e38

Recovery Note:

Same operation.
Different secrets.

The secret is hiding in plain sight.

This is what I got from using the wayback machine. The Recovery Fragment is the second part of the flag

Current: CYBERHEROCTF{m1sc_x0r_p4st3_n3v3r_

It said that the secret is hiding in plain sight so possibly 'AFTERIMAGE' could be the key to the new archived data. I used the same program but changed the data and the key.

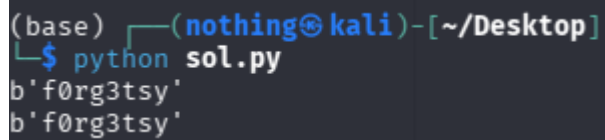
```
import binascii
```

```
hex_data = "27762622613d3e38"
```

```
# this converts the hex to byte
data = bytes.fromhex(hex_data)
key = b"AFTERIMAGE"
```

```
# this xor the the data that we get to the get first part of the flag
decoded = bytes(
    b ^ key[i % len(key)]
    for i, b in enumerate(data)
)
```

```
print(repr(decoded))
print(decoded)
```

A terminal window with a dark background. The prompt is (base) and the user is nothing@kali in the directory ~/Desktop. The command python sol.py has been executed, resulting in two lines of output: b'f0rg3tsy' and b'f0rg3tsy'.

```
(base) (nothing@kali)-[~/Desktop]
└─$ python sol.py
b'f0rg3tsy'
b'f0rg3tsy'
```

This is the last part of the flag then we construct everything together.

CYBERHEROCTF{m1sc_x0r_p4st3_n3v3r_forgets}